EDEXCEL INTERNATIONAL GCSE MATH (4MA1) • UNIT 7.01

Differentiation, Stationary Points & Kinematics

ANSWER KEY & SOLUTIONS

25 Structured Exam-Style Questions

Total Questions: 25 • Verified Step-by-Step Solutions

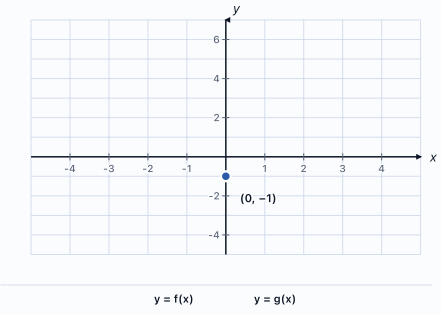
QUICK-REFERENCE ANSWER SUMMARY (25 ITEMS)

Q1 C	Q2 D	Q3 A	Q4 B	Q5 C
Q6 D	Q7 A	Q8 B	Q9 C	Q10 C
Q11 C	Q12 A	Q13 C	Q14 B	Q15 A
Q16 D	Q17 A	Q18 B	Q19 B	Q20 C
Q21 D	Q22 C	Q23 A	Q24 B	Q25 D

Differentiation, Stationary Points & Kinematics Practice Exercises

1 MCQ

The coordinate diagram shows the parabolic curve $f_1(x) = x^2 - 18x + 98$ in the domain of Differentiation, Stationary Points & Kinematics. By completing the square $f_1(x) = (x-h)^2 + k$, find the sum $k + f_1(11)$.



A 47

B 21

C 38 ✓

D 36

WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (Differentiation, Stationary Points & Kinematics):

1. Complete the square: $x^2 - 18x + 98 = (x-9)^2 - 81 + 98 = (x-9)^2 + 17$, giving minimum value $k = 17$. **[M1]**
2. Evaluate at $x = 11$: $f_1(11) = (2)^2 + 17 = 21$. **[M1]**
3. Sum the two quantities: $k + f_1(11) = 17 + 21 = 38$. **[A1]**

- **Final Verified Answer:** 38.

2 MCQ

A mathematical model in Differentiation, Stationary Points & Kinematics is defined by the cubic function $y_2(x) = 2x^3 - 9x^2 + 15x - 14$. Calculate the exact gradient of the tangent to the curve at the point where $x = 3$.

A 69

B 0

C $15 + 1$

D 15 ✓

WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (Differentiation, Stationary Points & Kinematics):

1. Differentiate term by term using the power rule: $\frac{dy}{dx} = 6x^2 - 18x + 15$. **[M1]**
2. Substitute $x = 3$: $\left.\frac{dy}{dx}\right|_{x=3} = 6(9) - 18(3) + 15 = 15$. **[M1][A1]**

- **Final Verified Answer:** 15.

3 MCQ

In Differentiation, Stationary Points & Kinematics, two displacement vectors are given by $\mathbf{u}_3 = \langle 9, -8 \rangle$ and $\mathbf{v}_3 = \langle 9, 7 \rangle$. Compute the exact scalar value of $\mathbf{u}_3 \cdot \mathbf{v}_3 + \|\mathbf{u}_3\|^2$.

A 170 ✓

B 25

C 145

D 282

WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (Differentiation, Stationary Points & Kinematics):

1. Compute the inner product: $\mathbf{u}_3 \cdot \mathbf{v}_3 = (9)(9) + (-8)(7) = 81 + (-56) = 25$. **[M1]**
2. Compute the squared norm: $\|\mathbf{u}_3\|^2 = 9^2 + (-8)^2 = 145$. **[M1]**
3. Combine: $25 + 145 = 170$. **[A1]**

- **Final Verified Answer:** 170.

4 MCQ

The linear growth profile shown in the diagram for Differentiation, Stationary Points & Kinematics follows an arithmetic progression with first term $a_1 = 10$ and common difference $d = 6$. Find the exact sum S_{15} of the first 15 terms.

A 94

B 780 ✓

C 870

D 1410

WORKED SOLUTION & VERIFICATION

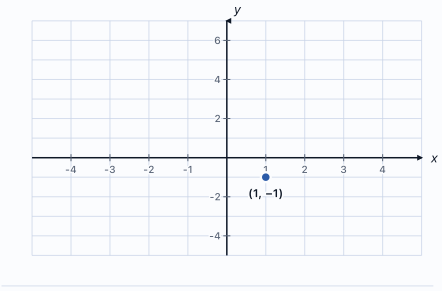
Mark Scheme & Examiner Solution (Differentiation, Stationary Points & Kinematics):

1. Use the sum formula for an arithmetic series: $S_n = \frac{n}{2}[2a_1 + (n-1)d]$. **[M1]**
2. Substitute $n = 15, a_1 = 10, d = 6$: $S_{15} = \frac{15}{2}[20 + 14 \times 6] = \frac{15}{2} \times 104 = 780$. **[M1][A1]**

- **Final Verified Answer:** 780.

5 MCQ

In Differentiation, Stationary Points & Kinematics, evaluate the exact definite integral $I_5 = \int_0^2 (18x^2 + 36x)\,\mathrm{d}x$.



A 144

B 156

C 120 ✓

D 61

WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (Differentiation, Stationary Points & Kinematics):

1. Integrate term by term: $\int (18x^2 + 36x)\,\mathrm{d}x = [6x^3 + 18x^2]_0^2$. **[M1]**
2. Substitute the upper and lower limits: $(6(2)^3 + 18(2)^2) - 0 = 48 + 72 = 120$. **[M1][A1]**

- **Final Verified Answer:** 120.

6 MCQ

In Differentiation, Stationary Points & Kinematics, consider the quadratic function $f_6(x) = x^2 - 20x + 189$. Express $f_6(x)$ in the completed-square form $(x-h)^2 + k$ and compute the exact value of $k + f_6(17)$.

A 237

B 138

C 220

D 227 ✓

WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (Differentiation, Stationary Points & Kinematics):

1. Complete the square: $x^2 - 20x + 189 = (x-10)^2 - 100 + 189 = (x-10)^2 + 89$, giving minimum value $k = 89$. **[M1]**
2. Evaluate at $x = 17$: $f_6(17) = (7)^2 + 89 = 138$. **[M1]**
3. Sum the two quantities: $k + f_6(17) = 89 + 138 = 227$. **[A1]**

- **Final Verified Answer:** 227.

7 MCQ

The graph illustrates the cubic model $y_7(x) = 2x^3 - 10x^2 + 33x - 21$ studied in Differentiation, Stationary Points & Kinematics. Find the instantaneous gradient $\frac{dy}{dx}$ of the tangent to the curve at $x = 8$.

- **A** 257 ✓
- **B** 417
- **C** 224
- **D** 61

WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (Differentiation, Stationary Points & Kinematics):

1. Differentiate term by term using the power rule: $\frac{dy}{dx} = 6x^2 - 20x + 33$. **[M1]**
2. Substitute $x = 8$: $\left.\frac{dy}{dx}\right|_{x=8} = 6(64) - 20(8) + 33 = 257$. **[M1][A1]**

- **Final Verified Answer:** 257.

8 MCQ

In Differentiation, Stationary Points & Kinematics, two displacement vectors are given by $\mathbf{u}_8 = \langle 10, -7 \rangle$ and $\mathbf{v}_8 = \langle 8, 5 \rangle$. Compute the exact scalar value of $\mathbf{u}_8 \cdot \mathbf{v}_8 + \|\mathbf{u}_8\|^2$.

- **A** 45
- **B** 194 ✓
- **C** 149
- **D** 264

WORKED SOLUTION & VERIFICATION

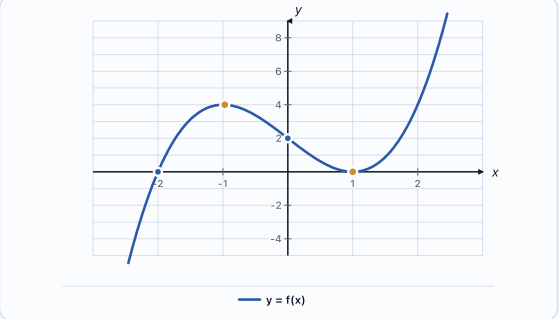
Mark Scheme & Examiner Solution (Differentiation, Stationary Points & Kinematics):

1. Compute the inner product: $\mathbf{u}_8 \cdot \mathbf{v}_8 = (10)(8) + (-7)(5) = 80 + (-35) = 45$. **[M1]**
2. Compute the squared norm: $\|\mathbf{u}_8\|^2 = 10^2 + (-7)^2 = 149$. **[M1]**
3. Combine: $45 + 149 = 194$. **[A1]**

- **Final Verified Answer:** 194.

9 MCQ

Find the coordinates of the local minimum stationary point on the curve $y = x^3 - 3x + 2$ for $x > 0$ (Curve #1).



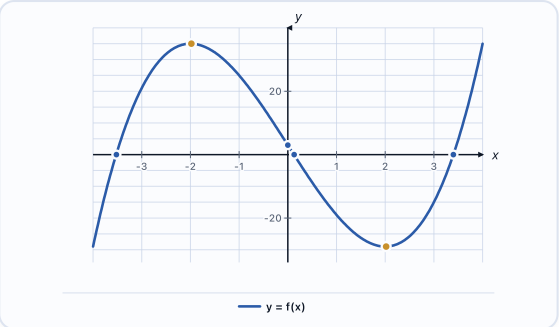
- **A** $(1, 2)$
- **B** $(2, -1)$
- **C** $(1, 0)$ ✓
- **D** $(-1, 0)$

WORKED SOLUTION & VERIFICATION

Differentiating gives $\frac{dy}{dx} = 3x^2 - 3 = 0 \implies x^2 = 1$. For $x > 0$, $x = 1$ with $\frac{d^2y}{dx^2} = 6 > 0$, and $y(1) = 0$.

10 MCQ

Find the coordinates of the local minimum stationary point on the curve $y = 2x^3 - 24x + 3$ for $x > 0$ (Curve #2).



A $(-2, -29)$

B $(2, 3)$

C $(2, -29)$ ✓

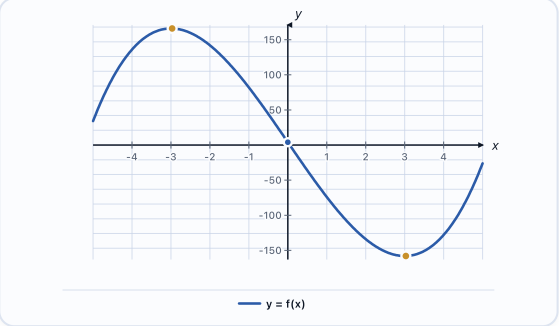
D $(4, -31)$

WORKED SOLUTION & VERIFICATION

Differentiating gives $\frac{dy}{dx} = 6x^2 - 24 = 0 \implies x^2 = 4$. For $x > 0$, $x = 2$ with $\frac{d^2y}{dx^2} = 24 > 0$, and $y(2) = -29$.

11 MCQ

Find the coordinates of the local minimum stationary point on the curve $y = 3x^3 - 81x + 4$ for $x > 0$ (Curve #3).



A $(6, -161)$

B $(-3, -158)$

C $(3, -158)$ ✓

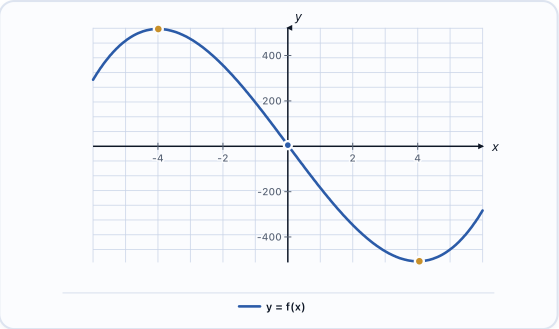
D $(3, 4)$

WORKED SOLUTION & VERIFICATION

Differentiating gives $\frac{dy}{dx} = 9x^2 - 81 = 0 \implies x^2 = 9$. For $x > 0$, $x = 3$ with $\frac{d^2y}{dx^2} = 54 > 0$, and $y(3) = -158$.

12 MCQ

Find the coordinates of the local minimum stationary point on the curve $y = 4x^3 - 192x + 5$ for $x > 0$ (Curve #4).



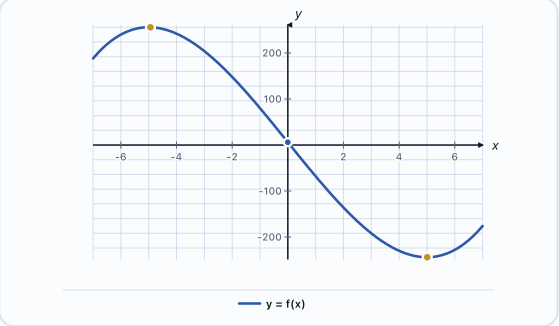
- **A** $(4, -507)$ ✓
- **B** $(8, -511)$
- **C** $(-4, -507)$
- **D** $(4, 5)$

WORKED SOLUTION & VERIFICATION

Differentiating gives $\frac{dy}{dx} = 12x^2 - 192 = 0 \implies x^2 = 16$. For $x > 0$, $x = 4$ with $\frac{d^2y}{dx^2} = 96 > 0$, and $y(4) = -507$.

13 MCQ

Find the coordinates of the local minimum stationary point on the curve $y = x^3 - 75x + 6$ for $x > 0$ (Curve #5).



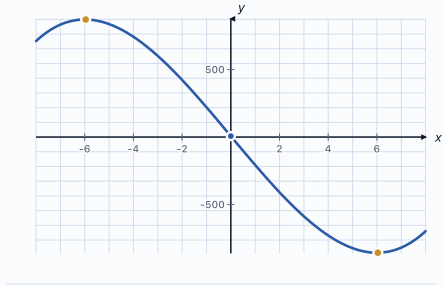
- **A** $(-5, -244)$
- **B** $(5, 6)$
- **C** $(5, -244)$ ✓
- **D** $(10, -245)$

WORKED SOLUTION & VERIFICATION

Differentiating gives $\frac{dy}{dx} = 3x^2 - 75 = 0 \implies x^2 = 25$. For $x > 0$, $x = 5$ with $\frac{d^2y}{dx^2} = 30 > 0$, and $y(5) = -244$.

14 MCQ

Find the coordinates of the local minimum stationary point on the curve $y = 2x^3 - 216x + 7$ for $x > 0$ (Curve #6).



A $(6, 7)$

B $(6, -857)$ ✓

C $(-6, -857)$

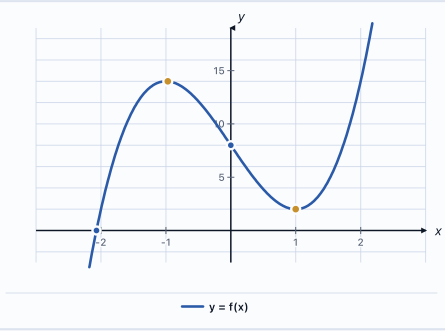
D $(12, -859)$

WORKED SOLUTION & VERIFICATION

Differentiating gives $\frac{dy}{dx} = 6x^2 - 216 = 0 \implies x^2 = 36$. For $x > 0$, $x = 6$ with $\frac{d^2y}{dx^2} = 72 > 0$, and $y(6) = -857$.

15 MCQ

Find the coordinates of the local minimum stationary point on the curve $y = 3x^3 - 9x + 8$ for $x > 0$ (Curve #7).



A $(1, 2)$ ✓

B $(1, 8)$

C $(-1, 2)$

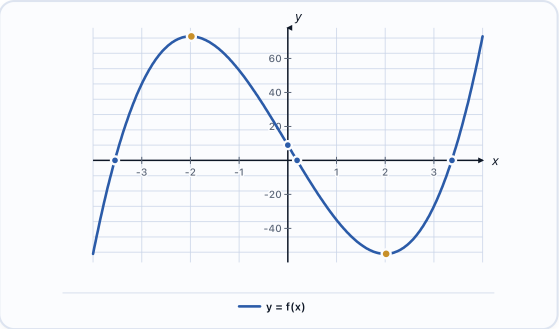
D $(2, -1)$

WORKED SOLUTION & VERIFICATION

Differentiating gives $\frac{dy}{dx} = 9x^2 - 9 = 0 \implies x^2 = 1$. For $x > 0$, $x = 1$ with $\frac{d^2y}{dx^2} = 18 > 0$, and $y(1) = 2$.

16 MCQ

Find the coordinates of the local minimum stationary point on the curve $y = 4x^3 - 48x + 9$ for $x > 0$ (Curve #8).



A $(-2, -55)$

B $(4, -59)$

C $(2, 9)$

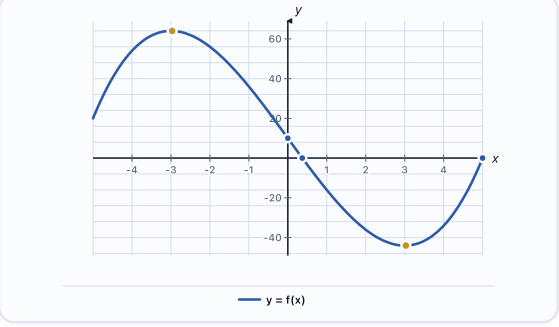
D $(2, -55)$ ✓

WORKED SOLUTION & VERIFICATION

Differentiating gives $\frac{dy}{dx} = 12x^2 - 48 = 0 \implies x^2 = 4$. For $x > 0$, $x = 2$ with $\frac{d^2y}{dx^2} = 48 > 0$, and $y(2) = -55$.

17 MCQ

Find the coordinates of the local minimum stationary point on the curve $y = x^3 - 27x + 10$ for $x > 0$ (Curve #9).



A $(3, -44)$ ✓

B $(3, 10)$

C $(6, -45)$

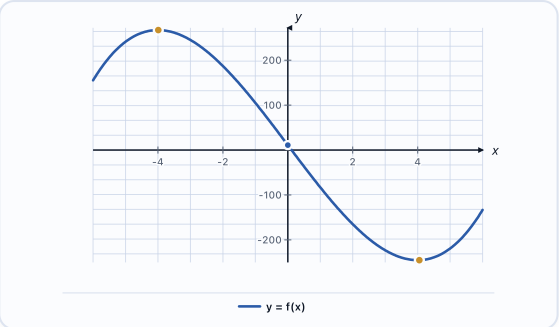
D $(-3, -44)$

WORKED SOLUTION & VERIFICATION

Differentiating gives $\frac{dy}{dx} = 3x^2 - 27 = 0 \implies x^2 = 9$. For $x > 0$, $x = 3$ with $\frac{d^2y}{dx^2} = 18 > 0$, and $y(3) = -44$.

18 MCQ

Find the coordinates of the local minimum stationary point on the curve $y = 2x^3 - 96x + 11$ for $x > 0$ (Curve #10).



A $(8, -247)$

B $(4, -245)$ ✓

C $(4, 11)$

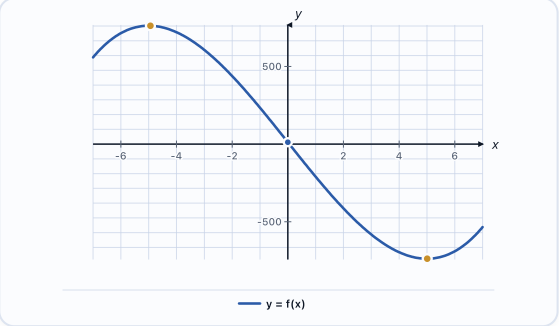
D $(-4, -245)$

WORKED SOLUTION & VERIFICATION

Differentiating gives $\frac{dy}{dx} = 6x^2 - 96 = 0 \implies x^2 = 16$. For $x > 0$, $x = 4$ with $\frac{d^2y}{dx^2} = 48 > 0$, and $y(4) = -245$.

19 MCQ

Find the coordinates of the local minimum stationary point on the curve $y = 3x^3 - 225x + 12$ for $x > 0$ (Curve #11).



A $(10, -741)$

B $(5, -738)$ ✓

C $(5, 12)$

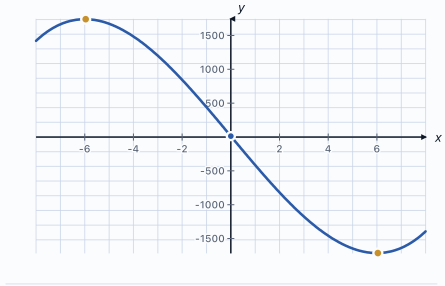
D $(-5, -738)$

WORKED SOLUTION & VERIFICATION

Differentiating gives $\frac{dy}{dx} = 9x^2 - 225 = 0 \implies x^2 = 25$. For $x > 0$, $x = 5$ with $\frac{d^2y}{dx^2} = 90 > 0$, and $y(5) = -738$.

20 MCQ

Find the coordinates of the local minimum stationary point on the curve $y = 4x^3 - 432x + 13$ for $x > 0$ (Curve #12).



A $(-6, -1715)$

B $(12, -1719)$

C $(6, -1715)$ ✓

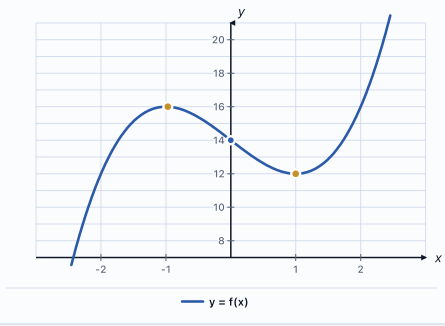
D $(6, 13)$

WORKED SOLUTION & VERIFICATION

Differentiating gives $\frac{dy}{dx} = 12x^2 - 432 = 0 \implies x^2 = 36$. For $x > 0$, $x = 6$ with $\frac{d^2y}{dx^2} = 144 > 0$, and $y(6) = -1715$.

21 MCQ

Find the coordinates of the local minimum stationary point on the curve $y = x^3 - 3x + 14$ for $x > 0$ (Curve #13).



A $(1, 14)$

B $(-1, 12)$

C $(2, 11)$

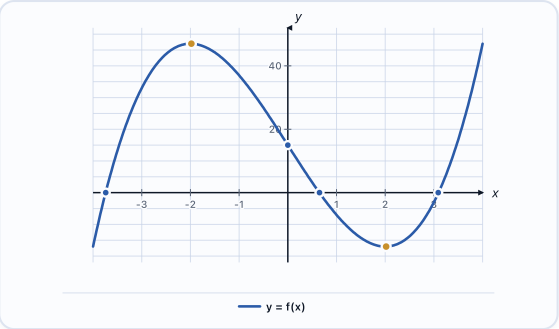
D $(1, 12)$ ✓

WORKED SOLUTION & VERIFICATION

Differentiating gives $\frac{dy}{dx} = 3x^2 - 3 = 0 \implies x^2 = 1$. For $x > 0$, $x = 1$ with $\frac{d^2y}{dx^2} = 6 > 0$, and $y(1) = 12$.

22 MCQ

Find the coordinates of the local minimum stationary point on the curve $y = 2x^3 - 24x + 15$ for $x > 0$ (Curve #14).



A $(-2, -17)$

B $(4, -19)$

C $(2, -17)$ ✓

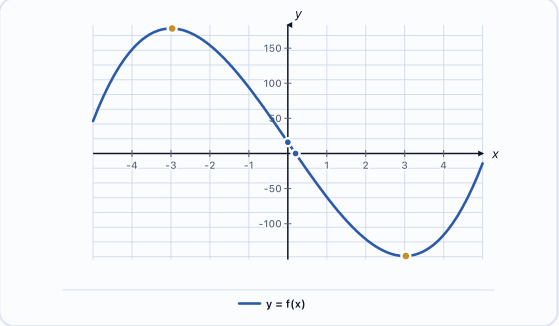
D $(2, 15)$

WORKED SOLUTION & VERIFICATION

Differentiating gives $\frac{dy}{dx} = 6x^2 - 24 = 0 \implies x^2 = 4$. For $x > 0$, $x = 2$ with $\frac{d^2y}{dx^2} = 24 > 0$, and $y(2) = -17$.

23 MCQ

Find the coordinates of the local minimum stationary point on the curve $y = 3x^3 - 81x + 16$ for $x > 0$ (Curve #15).



A $(3, -146)$ ✓

B $(-3, -146)$

C $(3, 16)$

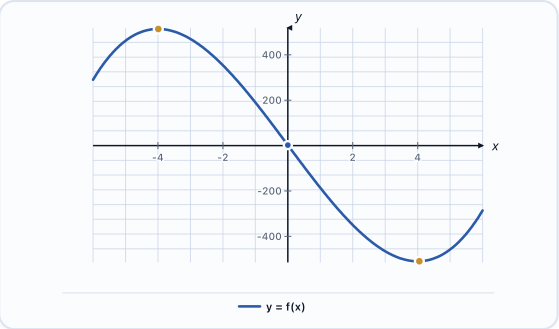
D $(6, -149)$

WORKED SOLUTION & VERIFICATION

Differentiating gives $\frac{dy}{dx} = 9x^2 - 81 = 0 \implies x^2 = 9$. For $x > 0$, $x = 3$ with $\frac{d^2y}{dx^2} = 54 > 0$, and $y(3) = -146$.

24 MCQ

Find the coordinates of the local minimum stationary point on the curve $y = 4x^3 - 192x + 2$ for $x > 0$ (Curve #16).



A $(4, 2)$

B $(4, -510)$ ✓

C $(-4, -510)$

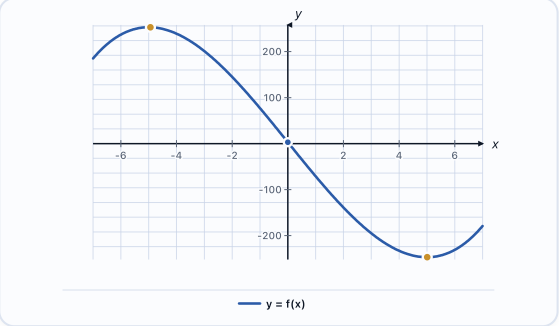
D $(8, -514)$

WORKED SOLUTION & VERIFICATION

Differentiating gives $\frac{dy}{dx} = 12x^2 - 192 = 0 \implies x^2 = 16$. For $x > 0$, $x = 4$ with $\frac{d^2y}{dx^2} = 96 > 0$, and $y(4) = -510$.

25 MCQ

Find the coordinates of the local minimum stationary point on the curve $y = x^3 - 75x + 3$ for $x > 0$ (Curve #17).



A $(10, -248)$

B $(5, 3)$

C $(-5, -247)$

D $(5, -247)$ ✓

WORKED SOLUTION & VERIFICATION

Differentiating gives $\frac{dy}{dx} = 3x^2 - 75 = 0 \implies x^2 = 25$. For $x > 0$, $x = 5$ with $\frac{d^2y}{dx^2} = 30 > 0$, and $y(5) = -247$.